YUMIN CHEN

SOFTWARE ENGINEER | AI ENGINEER

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PROFESSIONAL SUMMARY

Software Engineer and AI Engineer with experience building backend services, machine learning systems, LLM applications, and data-processing pipelines across autonomous-driving AI, telecommunications, and semiconductor manufacturing. Experienced in AI model deployment and optimization, RAG and AI-agent workflows, real-time data processing, and production services using Python, C++, and distributed technologies. Combines hands-on engineering with technical leadership and a focus on reliable, practical solutions.

TECHNICAL SKILLS

Programming	Python, C++, Go, Java, SQL, TypeScript
Machine Learning & AI	PyTorch, LLMs, RAG, AI agents, BERT, LSTM, Isolation Forest, ResNet, model inference
Model Optimization	INT8 quantization, CUDA, inference optimization, AI model deployment
Backend & APIs	FastAPI, REST APIs, Go and Java microservices
Data & Storage	Kafka, Flink, Apache Spark, Hive, SQL databases, data pipelines
DevOps & Tools	Docker, Kubernetes, Git, MLflow
Frontend	React

PROFESSIONAL EXPERIENCE

Independent Software Engineer | AI Engineer

Self-Employed | *February 2026 – Present* | *Wuhan, Hubei, China*

Working independently on software engineering and AI projects, with a focus on backend development, machine learning, and large language model (LLM) applications.

- Design and develop software solutions based on project requirements, focusing on maintainable code, practical functionality, and reliable performance.
- Build backend services and REST APIs using Python and related technologies, supporting application integration and data processing.
- Explore and develop AI-powered applications using large language models (LLMs), retrieval-augmented generation (RAG), and AI agent workflows.
- Work with machine learning models and AI frameworks to support model integration, inference, testing, and deployment.
- Develop data processing workflows and integrate databases, APIs, and external services to support end-to-end application functionality.
- Use Git, Docker, and modern development tools to organize code, manage changes, and improve development and deployment workflows.
- Research technical problems, evaluate suitable tools and frameworks, and implement solutions based on project requirements.

Senior AI Software Engineer

Black Sesame Technologies | *July 2022 – January 2026* | *Wuhan, Hubei, China*

Developed and optimized AI perception systems for autonomous driving and automotive AI chip platforms, focusing on model deployment, inference performance, and AI engineering tools.

- Developed and deployed 30+ PyTorch perception models for in-vehicle AI chips across five automotive customer programs.
- Optimized deep learning models through INT8 quantization, achieving up to 3.2× lower inference latency and 60% lower memory usage with less than 1% mAP loss.
- Developed and optimized C++ and CUDA kernels, improving inference throughput by 45%.
- Built a retrieval-augmented generation (RAG) assistant over 200,000+ pages of technical documentation, adopted by 600+ engineers and improving ticket resolution by 35%.
- Developed LLM-based tools for compiler-log analysis and troubleshooting, helping resolve approximately 40% of common model-conversion errors.
- Optimized LLM inference services using vLLM, Kubernetes, dynamic batching, and autoscaling, reducing GPU cost per query by 50%.
- Led a team of seven engineers and collaborated with automotive customers and Tier-1 suppliers on AI deployment and integration.

Software Engineer, AI Platform

FiberHome Technologies | *July 2020 – June 2022* | *Wuhan, Hubei, China*

Developed AI-driven operations and maintenance solutions for telecommunications infrastructure, combining machine learning, streaming data processing, and backend services.

- Built anomaly detection systems using LSTM and Isolation Forest to analyze more than two billion telemetry records per day.
- Developed predictive monitoring capabilities that identified potential link degradation approximately 30 minutes earlier and helped reduce outages by 25%.
- Implemented root-cause analysis workflows that reduced alarm noise by 70% and mean time to repair (MTTR) by 40%.
- Developed a BERT-based ticket classification and routing system that achieved 91% accuracy and saved an estimated 3,000 engineering hours annually.
- Built real-time data pipelines using Kafka and Flink, supporting prediction workflows with sub-second processing latency.
- Developed Go and Java microservices capable of handling more than 10,000 requests per second in Kubernetes environments.
- Implemented MLflow-based model lifecycle workflows, reducing the model release cycle from four weeks to one week.

Software Engineer

Yangtze Memory Technologies | *July 2018 – June 2020* | *Wuhan, Hubei, China*

Developed software and machine learning solutions for semiconductor manufacturing, focusing on wafer defect inspection, anomaly detection, manufacturing data processing, and engineering dashboards.

- Developed a ResNet-based wafer defect classification system that achieved 95% accuracy and automated approximately 60% of manual defect review.
- Implemented anomaly detection solutions to identify manufacturing irregularities and support process optimization, contributing to a 2% yield improvement.
- Developed Python and Java services for manufacturing execution system (MES) data processing, handling millions of records per day.
- Built data processing pipelines using Apache Spark and Hive to support manufacturing analytics and reporting.
- Developed React-based engineering dashboards used by more than 100 engineers to monitor manufacturing data and defect analysis results.

EDUCATION

Huazhong University of Science and Technology Wuchang Branch

Bachelor of Engineering | September 2014 – June 2018